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Legal Analysis of Forensic DNA Profiling: Issues and Challenges in India

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Abstract
Forensic DNA profiling has emerged as a crucial tool in modern criminal justice, offering high accuracy in identifying perpetrators and exonerating the innocent. This paper analyzes the legal and constitutional framework governing DNA evidence in India, particularly under the Bharatiya Nagarik Suraksha Sanhita, 2023 and the Bharatiya Sakshya Adhiniyam, 2023. Using a doctrinal and comparative approach, the study identifies a significant regulatory gap due to the absence of a dedicated DNA legislation.
The paper highlights key challenges including infrastructure deficits, weak chain of custody practices, and lack of standardized protocols. It further examines constitutional concerns relating to the right to privacy and protection against self-incrimination. The risk of data misuse, genetic surveillance, and absence of clear retention policies are also critically evaluated.
The study concludes that despite its scientific reliability, DNA profiling in India operates within an inadequate legal framework. It recommends the enactment of a comprehensive DNA regulatory law, improved forensic infrastructure, and stronger safeguards to ensure a balance between effective criminal investigation and protection of fundamental rights.

1. Introduction

Criminal justice administration has been greatly reshaped in the area of relying on the traditional testimonial evidence to the introduction of scientific and technological approaches. The most valid form of evidence in the past was ocular evidence especially eyewitness testimony. Nevertheless, the inherent weaknesses of human perception, memory failures and the increasing complexity of criminal operations has led to the necessity in moving to more objective and more reliable means of evidence. Forensic science in this regard has become an important instrument indeed with DNA profiling being referred to as the gold standard of forensic recognition.

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Deoxyribonucleic acid (DNA), which is the basic genetic material found in all living organisms, is a biological signature of an individual, except in the case of identical twins. DNA fingerprinting is also referred to as forensic DNA profiling which entails examination of certain parts of DNA, especially the Short Tandem Repeats (STRs), that are highly differentiated in different people. This scientific development has transformed the criminal investigations in that it has facilitated accurate identification of the culprits, connection of the suspects to the crime scenes, and innocence of the victims. The use of DNA technology in India started in the late 1980s and has continued to play a key role in the process of investigating major crimes such as homicide, sexual assault, and paternity cases.

The introduction of DNA profiling in the Indian judicial system has been slow and disjointed despite its scientific soundness. Traditionally, the courts used to refer to the provisions of Code of Criminal Procedure, 1973, especially Section 53 and 54 that allow medical examination of an accused. Nevertheless, the introduction of the Bharatiya Nagarik Suraksha Sanhita (BNSS), 2023, and the Bharatiya Sakshya Adhiniyam (BSA), 2023, has transformed the situation of institutionalizing forensic investigation. It is worth noting that BNSS Section 176 imposes the obligation of forensic scrutiny of those offending crimes that carry a sentence of imprisonment of seven years or above and hence puts a higher weight of scientific evidence in court to prosecute criminals.

Although this change in legislation is an indication of a progressive step towards evidence-based justice, it also creates a multifaceted constitutional and ethical issue. The mandatory character of collecting and examining biological samples conflicts with the basic rights, the right that is against self-incrimination in Article 20(3) and the right to privacy in Article 21 of the Constitution that was established in the historic case of the Justice K.S. Puttaswamy v. Union of India (2017). The problem is how to reconcile the interest of the state in quality law enforcement and the right to autonomy, bodily integrity, and genetic privacy of the person.

Furthermore, there is a massive gap in the legal system since there is no legislation that directly governs DNA profiling in India. With the withdrawal of the DNA Technology (Use and Application) Regulation Bill, 2019, many of the fundamental questions remain unanswered such as the creation of DNA data banks, collection and storage sample standards and protective measures against misuse of sensitive genetic information. This also creates an issue especially with the requirement of the new criminal laws that mandate forensic requirements that could overburden

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the available forensic infrastructure and raise the possibility of a procedural failure like contamination and misplaced chain of custody.

Moreover, there are also alerts connected with the protection of the data, genetic surveillance, and stigmatization of individuals, which makes the legal environment even more complicated. There are chances of abuse and infringement of the rights to privacy without clear guidelines on how to retain, access, and delete DNA profiles, especially when dealing with the people who are acquitted or wrongly charged.

It is on this basis that the current research paper aims to critically assess the legal, constitutional and procedural guidelines under which forensic DNA profiling in India is conducted. It attempts to address the question of whether the existing system sufficiently balances the achievement of scientific progress with the fundamental rights, and offers reforms that will ensure effective criminal investigation, as well as the preservation of civil liberties.

2. Literature Review

Increasing use of forensic DNA profiling within the criminal justice system has captured the interest of numerous scholars especially concerning its evidentiary power, legal framework and constitutional concerns. The literature on the topic widely acknowledges the fact that DNA profiling has become one of the most credible types of scientific evidence that can not only be used to convict, but also to acquit an innocent individual. Nevertheless, the Indian environment shows that the technological progress and the development of the legal system are not in line with each other, which is the complicated yet not completely formed legal system.

2.1 DNA profiling as a criminal justice tool

It is a common fact among scholars that the DNA profiling has revolutionized the way crimes are investigated in the sense that it offers a high level of accuracy in the identification of people. It is often referred to as the greatest evidentiary device that comes after fingerprints especially during sexual crimes, murder and paternity cases. Research indicates that it serves two purposes; not only in connecting the accused individuals with the crime scenes but also in acquitting the innocent people, which adds more weight to the fairness in the criminal justice system. Still, the credibility of DNA evidence depends on the correct gathering, conservation and examination, which is still the topic of concern in India.

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2.2 Current Legal and Institutional Environment.

One common idea (as expressed in the literature) is the lack of a specific legislative framework that would regulate DNA profiling in India. Although several efforts have been made to enforce all-encompassing bills, one of them being the DNA Technology (Use and Application) Regulation Bill, 2019, the regulatory provisions are still disjointed. At present, the rules of DNA evidence are regulated indirectly under the provisions of criminal procedure and evidentiary laws, including the Bharatiya Nagarik Suraksha Sanhita (BNSS), 2023 and the Bharatiya Sakshya Adhiniyam (BSA), 2023.

According to the scholars, as new legal changes broaden the boundaries of forensic examination, they do not introduce the detailed protection of the consent, data storage and control systems. There are also institutional issues that make the matter even more complicated, such as the shortage of funding of forensic labs, case backlog, and the inexistence of standardized procedures across jurisdictions. These shortcomings cast doubt on successful application of DNA technology and put the integrity of forensic results into doubt.

2.3 Evidentiary Value and Judicial Approach.

There is a conservative interpretation of DNA evidence by the judiciary in India. Although the generally accepted rules in courts are that DNA reports are admissible expert evidence the evidence is usually corroborative, but not conclusive. This conservative position can be explained by the fear of procedural issues, such as the inappropriate collection, contamination and discontinuation of the chain of custody.

Concurrently, the judicial rulings have come to acknowledge the probative power of the DNA profiling especially in sensitive cases like rape and paternity matters. Literature however shows that there are still inconsistencies in judicial reasoning, partly because of the lack of technical knowledge among legal practitioners and the legal judges. It is this field of forensic illiteracy that has resulted in over-trust as well as unnecessary distrust of DNA evidence.

2.4. Constitutional and Ethical Concerns.

Huge literature has been done on the constitutional effects of the DNA profiling especially when the protection against self-incrimination and right to privacy is taken into consideration. Since the historic case that declared the right to privacy as a core right, the question of whether requisite DNA sampling is legal or not has been acutely reviewed by scholars. The main issue is whether these practices pass the proportionality test particularly when weighed against the state interest in crime control.

Ethical issues also relate to the problem of bodily integrity, consent, and the misuse of genetic information. Lack of definite regulatory protection leads to the concerns of surveillance on the genetic level, illegal sharing of data, and discrimination. Another danger pointed out by scholars is a possibility of the so-called function creep, where the DNA data obtained due to the criminal investigation can be utilized in other areas, thus, compromising on the autonomy and dignity of the individual.

2.5 Procedural and Practice Problems.

Procedural flaws are always pointed out as a key constraint to the effective application of DNA evidence in India, based on empirical research. These are shortcomings of crime scene management, poor sample collection, delays during transportation, and absence of standard chain-of-custody guidelines. These discrepancies significantly diminish the evidential usefulness of the DNA profiling and can even result in a wrong conviction or acquittal.

Moreover, infrastructure bottlenecks in the criminal justice process include the lack of adequate forensic laboratories, lack of trained workers, and are caused by technological differences between regions. Such difficulties are especially acute taking into consideration compulsory forensic requirements in the new legal reforms that may turn the current system into its grave.

2.6 Research Gap

Although the available literature offers good information on the scientific, legal, and ethical aspects of DNA profiling, there are still number of gaps. The majority of the research focuses on these areas separately with little in the form of integration of constitutional analysis, the realities of the processes, and comparative standpoints. Moreover, without a specific DNA regulatory framework, there is a lack of thorough analysis of the consequences of the recent changes in the criminal law, specifically to the BNSS and the BSA.

This paper aims at filling these lapses by providing a comprehensive analysis that integrates the doctrinal, constitutional, and comparative analyses. It seeks to vigorously evaluate the appropriateness of the legal framework in India to manage the application of DNA profiling whilst protecting the basic rights.

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3. Research Methodology

The current research paper will use doctrinal and analytical research methodology to explore the legal, constitutional, and procedural framework of forensic DNA profiling in India. Since the research problem is a normative and interpretative question, the analysis is based on a qualitative account of legal texts, judicial decisions and the academic literature as the main points of analysis in the study.

3.1 Nature of Research

This study is doctrinal (black-letter law) in character, and deals with the systematic study of the provisions of statutes, the principles of the Constitution, and judicial interpretations of DNA profiling. It is also analytical as it critically assesses the sufficiency and competence of the current legal regimes especially following the recent reforms like the Bharatiya Nagarik Suraksha Sanhita, 2023 and the Bharatiya Sakshya Adhiniyam, 2023.

Also, the paper employs a comparative aspect, which refers to international jurisdiction, including the United Kingdom and the United States, to identify the best practices and regulatory frameworks.

3.2 Sources of Data

It is founded on the primary and secondary sources, which ensures the comprehensive and balanced analysis of the research.

A. Primary Sources

constitution of India (Articles 14, 20(3), and 21).

Statutory Framework:

: Bharatiya Nagarik Suraksha Sanhita (BNSS), 2023.

: Bharatiya Sakshya Adhiniyam (BSA), 2023.

: Bharatiya Nyaya Sanhita (BNS), 2023.

: Digital Personal Data Protection (DPDP) Act, 2023.

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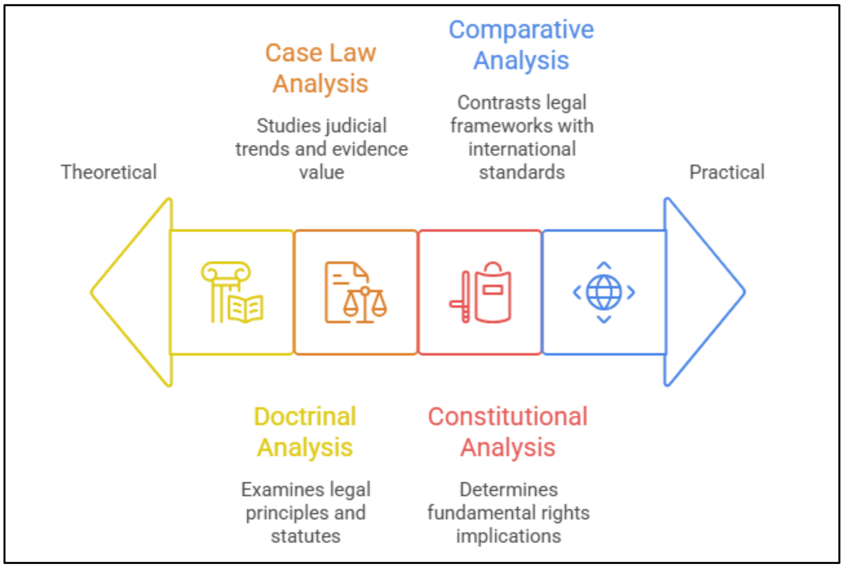

Figure 1. *Research methodology ranges from theoretical to practical application.*

3.3 Method of Analysis

The research has the following analytical tools used to meet its research goals:

: Doctrinal Analysis: To understand and judge the statutory provisions on DNA profiling according to the new criminal laws.

: Case Law Analysis: To study judicial trends regarding admissibility, reliability and evidences value of DNA evidence.

: Constitutional Analysis: To determine whether DNA profiling practices do not reflect the fundamental rights, especially the right to privacy and protection against self-incrimination.

: Comparative Analysis: To contrast the legal framework of India with international standards (e.g. UK and USA) to find out the gaps in regulations and best practices.

3.4 Scope of the Study

This study is limited by the scope of the legal and procedural issues that underlie forensic DNA profiling in India, and specifically, the scope is limited to:

: Criminal trials and investigations.

: DNA evidence, constitutional implications.

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: Data security and ethical issues.

: Effect of new criminal law reforms.

The limited comparative references to the foreign jurisdictions to contextualize the Indian framework are also included in the study.

3.5 Limitations of the Study

: The study is mostly doctrinal and does not involve the gathering of data empirically and in the field.

: Technicality The technicality of the DNA sequencing and the lab work is only addressed to the level required in legal analysis.

: Comparison is only done to a few jurisdictions and not all international practices.

: The fast changes in law and technology in the field of forensic science might have an impact on the sustainability of some findings.

3.6 Rationale for Methodology

The use of a doctrinal approach is justified because the study is aimed at examining legal principles, statutory interpretation, and constitutional validity and not experimental or empirical results. The comparative analysis also enhances the study due to the added advantage of giving a wider view of regulatory systems and best practices.

4. Legal Framework Analysis

The legal framework to forensic DNA profiling in India can be described as a fragmented and dynamic system with the statutory provisions, judicial interpretations, and recent criminal law reforms all controlling the usage of the same. Although DNA evidence is becoming more and more important in criminal cases, there is no specific and thorough law in India that regulates the usage, storage, and collection of genetic materials. The current legal framework is critically analyzed in this section, with its strengths and limitations and implications to the constitution.

4.1 Law According to DNA Profiling.

The legal position in pre-Reform (CrPC and Evidence Act) is 4.1.1.

Before the recent reforms in criminal law, in India, DNA profiling was indirectly regulated by the Code of Criminal Procedure, 1973 (CrPC) and the Indian Evidence Act, 1872. CrPC 53 and 54 gave medical practitioners the authority to examine the accused and the courts understood that it extended to the inclusion of biological samples like blood, semen, and hair.

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In the same vein, expert opinions such as forensic reports could be used in court under Section 45 of the Evidence Act. Nonetheless, these state provisions did not categorically address the issue of DNA profiling, so the judicial discretion and unequal use were observed. Lacking standardized procedures resulted in issues with admissibility and reliability of evidence most of the time.

4.1.2 Post-Reform Framework (BNSS and BSA, 2023)

The passage of the Bharatiya Nagarik Suraksha Sanhita (BNSS), 2023 and the Bharatiya Sakshya Adhiniyam (BSA), 2023 is a big move towards the institutionalization of forensic science in criminal justice.

Section 176 of the BNSS, 2023 requires the investigation of crimes with penalties of seven years or more by means of science, so the scientific evidence plays a bigger role in criminal proceedings. The BSA, 2023, supports the admissibility of expert evidence, such as the DNA report, as one of the elements of evidentiary evaluation.

Although these reforms are a step in the right direction of evidence-based justice, they are more oriented to increasing the application of forensic methods without specifying the procedure protection and controls over the use of DNA data.

4.1.3 Framework on Digital Personal Data Protection.

Digital Personal Data Protection (DPDP) Act, 2023 is one of the general laws on data protection in India. Yet, it is still unclear how it can be applied to genetic data. The DNA data is very sensitive and cannot be changed; thus, protection of such data should be more rigorous compared to common personal data.

The lack of clear statements dealing with:

: DNA data storage and retention.

: Consent mechanisms

: Certainly, access and sharing procedures.

invents a state of confusion about the safety of genetic privacy. Therefore, it cannot be sufficiently relied upon the general data protection law to govern the complexity of forensic DNA databases.

4.2 Lack of a DNA Regulatory Law.

One of the most important loopholes within the Indian legal system is the lack of a single piece of DNA legislation. The DNA Technology (Use and Application) Regulation Bill, 2019 that aimed at creating a National DNA Data Bank and a DNA Regulatory Board, was withdrawn, leaving a major legislative vacuum.

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This absence results in:

: Absence of uniform guidelines when collecting and analyzing.

: There are no strict guidelines of retaining or disregarding DNA profiles.

: Lack of independent checking systems.

There is an issue of the possible abuse, intrusion, and genetic surveillance without established regulation and enforced measures in the regulatory vacuum.

4.3 Interpretation and Evidentiary Value by the Court.

DNA evidence has been generally admitted by the Indian courts within the rubric of expert evidence. Judicial treatment, however, is a restrained and erratic treatment.

DNA evidence is often viewed as corroborative as opposed to conclusive especially where procedural irregularities are claimed.

Courts have put stress on the need to preserve the chain of custody in a bid to achieve evidentiary integrity.

It has been used in a number of cases in both convicting and acquitting innocent persons.

Simultaneously, lack of consistency in judicial reasoning may be due to:

: Absence of technical knowledge on DNA evidence.

: Differences in laboratory standards.

: Lapses in procedures during evidence processing.

This points to the necessity of homogenous judicial standards and forensic literacy among law-related stakeholders.

4.4 Constitutional Dimensions

DNA profiling brings out a serious constitutional issue especially in terms of basic rights.

4.4.1 Right to Privacy (Article 21)

Privacy as a recognized right has raised more questions on the activities of the state in terms of gathering and retention of individual data. In its turn, DNA profiling implies retrieving very sensitive genetic data, which brings up issues of informational privacy, bodily integrity, and autonomy.

The legality of such practices has to meet the test of proportionality, that:

: The measure has a valid purpose.

: It is reasonable and justified.

: There are sufficient safety measures.

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4.4.2 Right of Self-Incrimination (Article 20(3)).

Self-incrimination is questionable in regard to the mandatory use of biological samples. Although courts have made a distinction between testimonial evidence and physical evidence, the broadening of the usage of the new laws on forensic methods has created again the arguments over the boundaries of compulsion.

Researchers have claimed that in the absence of explicit statutory protection, mandatory DNA extraction could easily blur the lines between voluntary and involuntary evidence gathering.

4.5 Procedural and Institutional Issues.

Procedural and infrastructural limitations such as:

Animal weak chain of custody.

- : Possibility of contamination and intervention.
- : Absence of standard operating procedures.
- : The lack of qualified forensic professionals.
- : Delay in forensic laboratories.

All these contradict the credibility of DNA evidence and can result in miscarriage of justice regardless of the scientific validity of the technology.

4.6 Critical Evaluation

The existing legal system portrays a paradox, though India has adopted state-of-the-art forensic technology, India still depends on weak and scattered legal provisions. The decision to establish mandatory forensic investigation on the ground of BNSS is a positive move; still, the way this measure is implemented without an adequate regulatory framework can lead to the deterioration of procedural fairness and constitutional rights.

Lack of a specific DNA law, lack of data protection rights and infrastructural limitations demonstrate the necessity of the comprehensive and rights-based legal framework.

5. Comparative Analysis

Forensic DNA profiling laws differ widely among jurisdictions and some nations have established a high level of legal regulation and institutional protection against abuse, including the United Kingdom and the United States. India, on the contrary, still works in a piece meal system of law with no special law on DNA. Comparative analysis assists in establishing the best practices and regulatory loopholes, which can provide useful information in reform.

5.1 United Kingdom

The United Kingdom boasts of one of the most developed and controlled DNA profiling systems in the globe. Police and Criminal Evidence Act (PACE), 1984 and its amendments give a clear statutory basis of collecting and using DNA samples. The UK has also the National DNA Database (NDNAD) that is one of the largest DNA databases maintained in the world in terms of forensic DNA.

Key features include:

: Lawful control of the DNA collection and storing.

: Strict separation of suspects, convicts and volunteers.

: Good monitoring systems and autonomous officials.

: Judicial review and adherence to human rights (ECHR)

Notably, the UK framework consists of stringent rules of retention and deletion, especially where the judiciary has interfered in such cases as S and Marper v. United Kingdom, who underlined the right to privacy.

5.2 United States

DNA profiling in the US is administered under the DNA Identification Act, 1994, and this gave rise to the Combin-DNA Index System (CODIS). The system is deployed at the federal, state, and local levels, which makes it very broad and help to share the data efficiently.

Key features include:

: State level and federal statutory framework.

: DNA database (offers, arrested people, missing persons) classification.

: Uniform forensic practices and accreditation standards.

: High priority on efficiency in law enforcement.

Nevertheless, it has been questioned how the databases of DNA can be expanded and that it might also be invading the privacy of the people especially in the case of the arrestees and other potential offenders who are not convicted.

5.3 India

The process of DNA profiling in India is still lawfully underdeveloped and disjointed. There is no specific law governing the collection, storage, or databanking of DNA, although recent reforms like the Bharatiya Nagarik Suraksha Sanhita (BNSS), 2023 require the use of forensic investigation.

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Key issues include:

: Lack of a thorough DNA regulatory statute.

: Deficiency of standard procedures in collection and analysis.

: Poor chain-of-custody systems.

: Weak forensic facilities and capacity.

Table 1: Comparative Table: DNA Profiling Framework (India vs UK vs USA)

Aspect	**India**	**United Kingdom**	**United States**
Legal Framework	No dedicated DNA law; governed by BNSS, BSA	Comprehensive statutory framework (PACE Act)	DNA Identification Act, 1994
DNA Database	No national DNA database (proposed but not enacted)	National DNA Database (NDNAD)	CODIS (national + state databases)
Regulatory Authority	No independent regulatory body	Independent oversight bodies	FBI oversight + federal/state agencies
Collection of Samples	Mandated under BNSS for serious offences	Clearly regulated with safeguards	Broad collection, including arrestees
Retention & Deletion Policy	No clear statutory rules	Strict retention limits (post Marper case)	Varies by state; often expansive
Privacy Protection	Limited; governed indirectly by DPDP Act	Strong privacy safeguards under ECHR	Moderate; concerns over expansion
Chain of Custody Standards	Weak and inconsistent	Highly standardized protocols	Standardized and regulated
Infrastructure & Capacity	Limited labs and backlog issues	Advanced and well-equipped	Highly developed forensic system
Judicial Approach	Cautious; often corroborative evidence	Strong reliance with safeguards	Widely accepted and relied upon

Incompetent safeguards of data protection: The cancellation of the DNA Technology (Use and Application) Regulation Bill, 2019 has further polarized the gap in the regulation, as it does not cover some of the key areas of DNA regulation.

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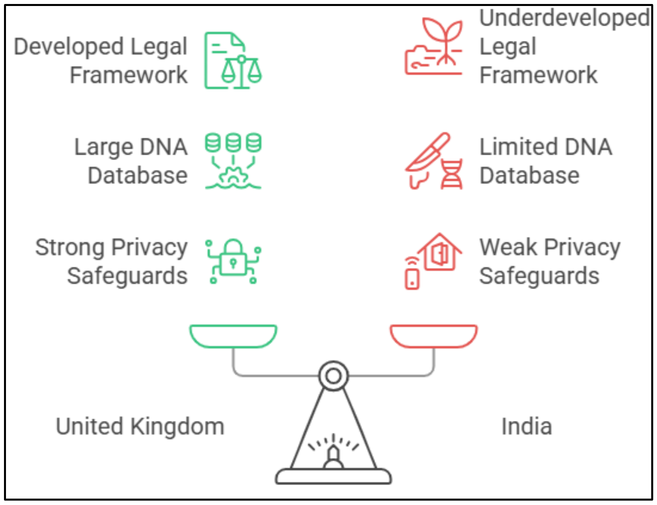

Figure 2. Comparing DNA Profiling Systems

5.5 Key Comparative Insights

Regulatory Gap in India: There is no clear target statutory and institutional framework in India, as in the case of the UK and the USA, and this uncertainty is attributed to the use of DNA evidence.

- Privacy Protections: UK has the best example of balancing the rights of investigations and privacy, whereas the US values the efficiency of enforcing the rights.
- Standardization: There are homogeneous protocols and accreditation systems in both UK and USA, which are not very prevalent in India.
- Data Protection: India does not have sufficient data protection law in comparison with specific DNA rules in other countries.
- Institutional Capacity: Developed jurisdictions possess a powerful infrastructure whereas India is characterized with heavy resource constraints.

6. Results and Recommendations.

6.1 Findings

- Legislative void: India does not have a wholesome law on DNA and so there is disjointed regulation.
- Infrastructure Deficit: There are no sufficient labs or trained personnel to do mandated forensic use under BNSS.
- Constitutional Issues: DNA collection is a matter that concerns Article 21 (privacy) and Article 20(3) (self-incrimination).

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- : Procedural Weaknesses: There are evidentiary weaknesses in that there is chain of custody violation and contamination.

Judicial Inconsistency: DNA evidence can be frequently considered to be corroborative because of procedural lapses.

Others: Data Protection Risks: There are no established regulations on storage and erasure that can result into possible abuse and genetic surveillance.

6.2 Recommendations

- : Enact DNA Law: Pass a statute that is all-encompassing in terms of collection, storage and databanks.
- : Enhance Privacy Protection: Consent, Retention and Right to be Forgotten.
- : Standardize Procedures: Use standard SOPs of collection and chain of custody.
- : Improve Infrastructure: Increase forensic laboratories and decrease backlog.
- : Capacity Building: Educate police, prosecutors and judges on forensic science.
- : Judicial Guidelines: define the admissibility and the evidentiary value of DNA evidence.
- : Self-Governing Regulation: Have DNA databanks regulated.

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